

RL-preview-hw4

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1 Problem 1

Interpretation 1: Approximate Q-VI in learning setting

From Bellman Optimal Equation:

$$Q_{k+1}(s, a) \leftarrow R_a^s + \gamma \sum_{s' \in S} P_{ss'}^a (\max_{a' \in A} Q_k(s, a'))$$

, and we estimated reward and transition probability from sampling, but we can't just assigning the left value to the right, because we use sampling to get the reward and transition probability value, it have the high variance, so we use moving average method to update the new Q value, the moving average method is like: new estimate = (1- α)*old estimate + α *new sample, so the first interpretation is:

$$Q(s_t, a_t) \leftarrow (1-\alpha)Q(s_t, a_t) + \alpha(r_{t+1} + \gamma \max_{a' \in A} Q(s_{t+1}, a'))$$

Interpretation 2: Correction by Bellman error

Bellman error is defined as $\delta = (r(s, a) + \gamma \max_{a' \in A} Q(s', a')) - Q(s, a)$, target is $(r(s, a) + \gamma \max_{a' \in A} Q(s', a'))$, we use this to update the original Q value by using learning rate α , so the second interpretation is:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_{t+1} + \gamma \max_{a' \in A} Q(s_{t+1}, a') - Q(s_t, a_t))$$

The reason why Q-learning is off-policy learning, is that we want to learn a target policy $\pi(a|s)$ that makes $Q^\pi(s, a) = Q^*(s, a)$, but in the meantime we interact with environment by using another policy called behavior policy (like ϵ -greedy exploration etc.), since the policy we learn isn't the policy we interact with the environment, so the Q-learning is off-policy learning.

2 Problem 2

The key idea of the design in the loss function is that DQN loss function want to minimize the bellman error to make the Q-network satisfy the Bellman Optimal Equation. The loss function is

$$L(w) := \frac{1}{2} E_{(s, a, r, s' \sim \rho)} [(r + \gamma \max_{a' \in A} \hat{Q}(s', a'; w) - \hat{Q}(s, a'w)]$$

- (s,a;w) is the current Q-value estimated by Q-network
- $r + \gamma \max_{a' \in A} \hat{Q}(s', a'; w)$ is the bellman target, consisting the immediate reward and the estimated optimal return.
- The difference is called Bellman error, which we want to minimize.
- The expectation averages the error over sampled transition.

The principle behind the design is that DQN transform the reinforcement learning problem into the supervised regression problem, where we look the bellman target as the labeled value for training the Q-network.

3 Problem 3

The Noisy net is a noisy linear layer that combines a deterministic and noisy stream. Over time, the network can learn to ignore the noisy stream, but will do so at different rates in different parts of the state space, allowing state-conditional exploration with a form of self-annealing.